IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2018

EEE PART III/IV: MEng, BEng and ACGI

Corrected copy

OPTOELECTRONICS

Wednesday, 5 December 9:00 am

Time allowed: 3:00 hours

There are SIX questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : R.R.A. Syms
Second Marker(s) : O. Sydoruk

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Fundamental constants

$e = 1.6 \times 10^{-19}$ C

$m_e = 9.1 \times 10^{-31}$ kg

$\varepsilon_0 = 8.85 \times 10^{-12}$ F/m

$\mu_0 = 4\pi \times 10^{-7}$ m kg/C^2

$c = 3 \times 10^8$ m/s

$h = 6.62 \times 10^{-34}$ Js

Maxwell's equations – integral form

$\iint_A \underline{\mathbf{D}} \cdot d\underline{a} = \iiint_V \rho \, dv$

$\iint_A \underline{\mathbf{B}} \cdot d\underline{a} = 0$

$\int_L \underline{\mathbf{E}} \cdot d\underline{L} = -\iint_A \partial\underline{\mathbf{B}}/\partial t \cdot d\underline{a}$

$\int_L \underline{\mathbf{H}} \cdot d\underline{L} = \iint_A [\underline{\mathbf{J}} + \partial\underline{\mathbf{D}}/\partial t] \cdot d\underline{a}$

Maxwell's equations – differential form

$\text{div}(\underline{\mathbf{D}}) = \rho$

$\text{div}(\underline{\mathbf{B}}) = 0$

$\text{curl}(\underline{\mathbf{E}}) = -\partial\underline{\mathbf{B}}/\partial t$

$\text{curl}(\underline{\mathbf{H}}) = \underline{\mathbf{J}} + \partial\underline{\mathbf{D}}/\partial t$

Material or constitutive equations

$\underline{\mathbf{J}} = \sigma \underline{\mathbf{E}}$

$\underline{\mathbf{D}} = \varepsilon \underline{\mathbf{E}}$

$\underline{\mathbf{B}} = \mu \underline{\mathbf{H}}$

Vector calculus (Cartesian co-ordinates)

$\text{grad}(\phi) = \partial\phi/\partial x \, \underline{i} + \partial\phi/\partial y \, \underline{j} + \partial\phi/\partial z \, \underline{k}$

$\text{div}(\underline{\mathbf{F}}) = \partial F_x/\partial x + \partial F_y/\partial y + \partial F_z/\partial z$

$\text{curl}(\underline{\mathbf{F}}) = \underline{i} \{\partial F_z/\partial y - \partial F_y/\partial z\} + \underline{j} \{\partial F_x/\partial z - \partial F_z/\partial x\} + \underline{k} \{\partial F_y/\partial x - \partial F_x/\partial y\}$

$\text{curl} \{\text{curl}(\underline{\mathbf{F}})\} = \text{grad} \{\text{div}(\underline{\mathbf{F}})\} - \nabla^2 \underline{\mathbf{F}}$

$\iint_A \underline{\mathbf{F}} \cdot d\underline{a} = \iiint_V \text{div}(\underline{\mathbf{F}}) \, dv$

$\int_L \underline{\mathbf{F}} \cdot d\underline{L} = \iint_A \text{curl}(\underline{\mathbf{F}}) \cdot d\underline{a}$

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1. a) State and explain Snell's law. A light beam incident from air onto water at an angle of 30° to the surface normal is seen to refract at 22.1°. Calculate the refractive index of water.

[4]

b) An underwater light source emits light equally in all directions. Explain the physical factors limiting the fraction of the emitted power that can escape from the water surface.

[4]

c) Light is generated by electroluminescence in a semiconducting slab of refractive index 3.5, with an air surround. Calculate the critical angle and the reflection coefficient at normal incidence.

[4]

d) Using simple approximations, estimate the fraction of power that can escape from the semiconductor surface.

[6]

e) Explain briefly the implications of your calculations above for i) light emitting diodes in communication systems, and ii) underwater swimming pool lighting.

[2]

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2. The eigenvalue equations at wavelength λ for a symmetric dielectric waveguide of height h formed using materials of refractive indices n_1 and n_2, with $n_1 > n_2$, are:

$$\tan(\kappa h/2) = \gamma/\kappa \quad \text{and} \quad \tan(\kappa h/2) = -\kappa/\gamma$$

Here $\kappa = \sqrt{(n_1^2 k_0^2 - \beta^2)}$, $\gamma = \sqrt{(\beta^2 - n_2^2 k_0^2)}$, β is the propagation constant and $k_0 = 2\pi/\lambda$.

a) Sketch the structure and show geometrically how κ and β are related.

[5]

b) Show that the values of β, κ and γ at cutoff are $k_0 n_2$, $k_0\sqrt{(n_1^2 - n_2^2)}$ and zero, respectively.

[5]

c) Using the expressions above, derive the condition for single-mode operation of the guide.

[6]

d) A symmetric dielectric waveguide is formed using glasses of refractive indices of 1.51 and 1.5. How should the core thickness h be chosen for single-mode operation at 1.5 μm wavelength? How should n_1 be altered, to increase this value to 6 μm?

[4]

3. a) Figures 3.1 i) - iv) show different channel waveguide circuits based on Y-junctions. Assuming perfect symmetry, and explaining your reasoning, calculate the guided output amplitudes in each case, for the input amplitudes shown.

[10]

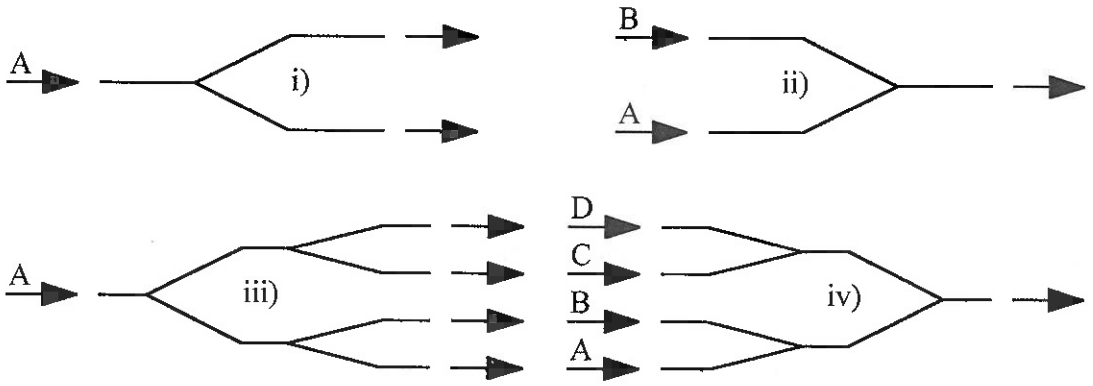


Figure 3.1

b) Figure 3.2 shows a channel guide circuit equipped with phase modulators driven from a common voltage source to apply the phase shifts shown. Calculate the output amplitudes A_1 and A_2, and sketch the variations of the corresponding powers with Φ.

[6]

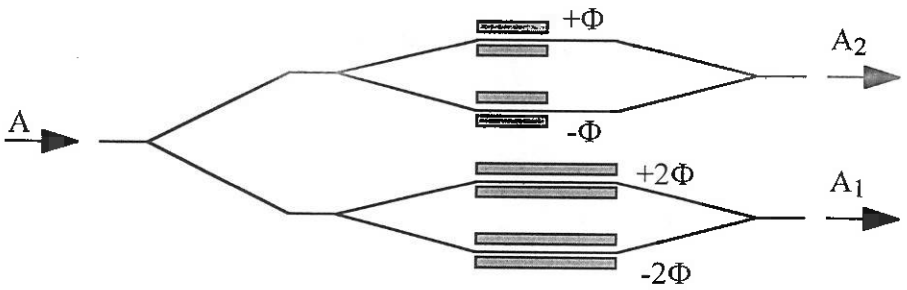


Figure 3.2

c) Propose and explain a practical application of the circuit in b).

[4]

4. a) The Fermi-Dirac function is $F(E) = 1/\{1 + \exp[(E - E_F)/KT]\}$, where E is the electron energy, E_F is the Fermi level, K is Boltzmann's constant and T is absolute temperature. What does F(E) represent? What is the meaning of E_F? Sketch the variation in F(E) with energy, for two representative values of T, both with $T > 0$.

[4]

b) For electrons, the density of states function is $S(E) = C_e \sqrt{(E - E_C)}$ for $E \geq E_C$, where E_C is the energy at the conduction band edge and C_e is a constant. Similarly, for holes, $S(E) = C_h \sqrt{(E_V - E)}$ for $E \leq E_V$, where E_V is the energy at the valence band edge and C_h is a constant. Sketch the variations in the density of states with energy, on either side of the forbidden band.

[4]

c) Propose approximations to F(E) valid for $E >> E_F$ and to 1 - F(E) valid for $E << E_F$. Using these, sketch the electron distribution n(E) at the bottom of the conduction band and the hole distribution p(E) at the top of the valence band.

[6]

d) Using your answer to c), explain why optical absorption is weak for photon energies close to the energy gap $E_g = E_C - E_V$, and strong for energies much larger than E_g.

[6]

5. a) Explain the meaning of quantum efficiency as applied to i) lasers and ii) photodiodes. What are the main factors limiting quantum efficiency in each case?

[5]

b) Using band diagrams where necessary, explain the steps that can be taken to maximise quantum efficiency in lasers.

[5]

c) Using band diagrams where necessary, explain the steps that can be taken to maximise quantum efficiency in photodiodes.

[5]

d) A student's logbook was spoilt when a cup of coffee was spilled over it during his summer placement. He thinks he can read the measurements of silicon photodiode performance shown in Table 5.1. Based on the quantum efficiency, which of the readings do you think have been corrupted, and which are valid? The energy gap of silicon is $E_g = 1.12$ eV.

[5]

Wavelength (μm)	Responsivity (A/W)
0.7	0.378
0.8	0.483
0.9	0.761
1.0	0.322
1.2	0.193

Table 5.1

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6. a) By considering the round trip up and down a laser cavity with propagation constant β, gain g, end-mirror reflectivities R_1 and R_2 and length L, derive the overall condition for laser oscillation. Divide this into separate conditions on phase and gain. Sketch the variation in wave amplitude as it propagates up and down the cavity.

[7]

b) Ignoring the gain, sketch the variation of power with time for a pulse of photons injected into the cavity. Hence, derive an expression for the photon lifetime.

[5]

c) A semiconductor laser is 250 μm long, and the lasing stripe has effective index 3.5. Calculate the spectral separation of adjacent modes near the peak emission at 1.5 μm wavelength, and the gain needed to reach threshold. Calculate the photon lifetime.

[8]